

Music and multimedia streaming over the Internet - Lab 4

your names here

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4 Deliveries

4.1 Ping

4.1.1 To known IP address

1. Chosen IP:
2. Exact command issued:
3. List of protocols involved:

Choose an Echo Request or an Echo Reply packet and show:

1. Packet type (Request or Reply):
2. Source IP address:
3. Destination IP address:

4.1.2 To named host

1. Chosen host name:
2. Exact command issued:
3. IP address of the chosen host:
4. Describe the additional packet types observed with respect to exercise 4.1.1:

4.2 Netcat

4.2.1 TCP

1. Report the commands used to run both the Netcat client and the Netcat server
 - Server:
 - Client:
2. Identify the packets used to initiate and close the TCP connection. Provide a screenshot or a list of them:
3. Write here the source port of the packets going in the direction client $\rightarrow$ server:

4.2.2 UDP

1. Report the commands used to run both the Netcat client and the Netcat server
 - Server:
 - Client:
2. What is the main difference in the number of packets when using UDP instead of TCP? Which packets are new, and which ones are missing?
3. What are the differences between a TCP and a UDP packet? Could you briefly explain why?
4. Do you notice any strange behavior in the order of the packets?

4.3 Web

4.3.1 HTTP Server

1. Briefly report the type of packets observed in Wireshark
2. Could you find the username and password typed in the browser among the packets?
3. If so, where? If not, why?

4.3.2 HTTPs Server

1. Briefly report the type of packets observed in Wireshark
2. Could you find the username and password typed in the browser among the packets?
3. If so, where? If not, why?